Macroeconomic Theory Qualifying Exam

The Playbook

A Pattern-Based Reference for Exam-Day Triage and Solution

ASU Department of Economics
Compiled from sample qualifiers, 2016–2024

How to use this document.

This is a *reference*, not a textbook. Three intended uses:

1. **During practice exams.** Open to Part I (Triage) when you read a new problem. Open to Part III (Recipes) when you reach the closed-form computation step. Open to Part IV (Proof Techniques) when you face a "show that" question.
2. **During post-mortem.** After grading a practice exam, locate the relevant pattern here, see what move you missed, and feed the lesson back into Part VIII (Annotated Index).
3. **In the last 48 hours.** Skim Part I, Part III closed-form table, and Part VII (Pitfalls). Don't try to absorb anything new.

Scope: content is derived purely from the pattern of nine sample exams (June 2016 through June 2024). It is *not* a substitute for lecture notes; it is a complement to them, optimized for the exam.

Contents

1 Pattern Overview: What the Exam Tests

Across nine sample qualifiers (2016–2024) covering 29 problems, the distribution and recurrence is:

Topic	**Frequency**	**Notes**
Overlapping Generations (OLG)	9 of 29 ($\sim$31%)	Appears *every* year 2018–2024
Growth / Dynamic Programming	12 of 29 ($\sim$41%)	Either planner or decision problem
Asset Pricing (Lucas tree)	6 of 29 ($\sim$21%)	Recurs roughly every other year
Search (McCall)	2 of 29 ($\sim$7%)	Swing topic; 2020, 2022

Two important caveats. First, "Growth/DP" is heterogeneous: it ranges from a textbook planner problem (2016 Q1) to highly novel decision problems with custom state spaces (2018 Q3 resource extraction; 2019 Q3 irreversible employment; 2022 Q1 project choice; 2023 Q3 alternating management). Galina's signature appears here. Second, RCE *definition* is asked in roughly half of all problems regardless of topic; treat it as a mandatory skill, not a topic.

The Recurring Five-Step Structural Template

Across the topics, problem structures are remarkably uniform. **Roughly 80% of problems follow this five-step template:**

1. **Set up** the problem (recursively or sequentially). Identify state variables, choice variables, constraints.
2. **Derive** FOCs, the Envelope condition, and combine to obtain the (stochastic) Euler equation.
3. **Define** the equilibrium concept (RCE, competitive with sequence of markets, or Arrow-Debreu).
4. **Specialize** to closed form. The standard simplifications are log utility, Cobb-Douglas production, full depreciation $\delta = 1$, and (when stochastic) iid shocks.
5. **Compare** (comparative statics) or evaluate a fictional economist's claim.

Steps 1, 2, 4 reward speed and pattern recognition. Steps 3 and 5 reward precision and conceptual clarity. The graders care most about: (i) precise equilibrium definitions, (ii) correctly handling endogenous price feedback in claims, (iii) clean derivations.

The "Economist X" Trap

At least 11 of the 29 problems include a "Economist X claims Y — is she right?" subpart (2016 Q2; 2017 Q1d, Q3d; 2019 Q2d; 2020 Q1e; 2021 Q1d, Q2d; 2022 Q2c; 2023 Q1d; 2024 Q2c, plus implicit instances). **The claim is almost always wrong, or right for the wrong reason.** The trap mechanic is catalogued in Part 7. If you see this subpart, do not skim — it is testing a specific GE feedback the claim ignores.

2 Part I — Triage and Recognition

When you read a new problem, the cost of misidentifying its category is large. This section gives you the lookup that fires *before* you start writing.

2.1 Linguistic Triggers: Mapping Phrases to Frameworks

The exam writers signal the framework with consistent phrases. The following table is the recommended first lookup.

If the problem says . . .	**Framework**	**Past instances**
"Individuals live for two periods"	Two-period OLG	2016 Q2, 2019 Q2, 2020 Q3, 2021 Q1, 2022 Q2, 2023 Q1, 2024 Q2
"Individuals live for three periods" / mortality / death probability	Three-period OLG	2018 Q2
"A cohort of size L (or N) is born every period"	OLG (look for capital, growth)	2016 Q2, 2017 Q3, 2019 Q2
"Tree(s) yielding dividends z_t" / "infinite-lived agent" / "mass one of identical agents" valuing $\sum \beta^t u(c_t)$ with stochastic dividends	Lucas tree asset pricing	2016 Q3, 2017 Q1, 2018 Q1, 2019 Q1, 2021 Q2, 2023 Q2
"Wage offer drawn from $F(w)$" / "unemployed worker draws an offer" / "reservation wage"	McCall search	2020 Q2, 2022 Q3
"Linear production technology $f(k) = Ak$"	AK / endogenous growth	2017 Q2, 2023 Q3, 2024 Q1
"Aggregate capital affects productivity $A_t = \int k_{it}\, di$"	Endogenous-growth externality	2017 Q2
"Government taxes τ_t and runs a balanced budget"	Ramsey-style fiscal problem	2016 Q4, 2017 Q4, 2021 Q2, 2024 Q3
"Households value consumption *and* leisure" / "$h_t + l_t \leq 1$"	Growth with endogenous labor	2016 Q4, 2020 Q1
"Home-produced goods" / "$c_2 = G(l)$ or Ah"	OLG with home production	2022 Q2, 2024 Q2
"Two types of individuals" / "fraction π" / "skilled vs. unskilled"	OLG with heterogeneity	2016 Q2, 2019 Q2, 2022 Q2, 2023 Q1
"Disaster wipes out fraction of capital with probability q"	Stochastic AK / disaster risk	2024 Q1
"Choose once, irreversibly" / "project choice" / "employment choice"	Decision problem with cutoff rule	2019 Q3, 2022 Q1
"Continuum of individuals" / "each chooses to be a worker or entrepreneur"	Heterogeneous-agent RCE with occupational choice	2024 Q3
"Skill accumulation" / $s_{t+1} = (1-\delta)s_t + As_t^{1-\gamma} i_t^{\gamma}$	Decision problem with nonlinear LoM	2021 Q3

If the problem says ...	Framework	Past instances
"Two countries / two trees / international bond"	Multi-asset pricing	2017 Q1, 2019 Q1
"Time-varying net supply" / "external investors demand $\hat{y}_t$"	Asset pricing with supply shocks	2018 Q1
"Resource grows at rate g" / "unused stock grows"	Resource extraction (custom LoM)	2018 Q3
"Alternating management" / "two agents share an asset"	Strategic dynamic problem	2023 Q3

2.2 The 5-Step Structural Template (Master Procedure)

The vast majority of problems decompose into the same five steps, even when the framework changes. Internalize this template; it lets you start writing within 60 seconds.

The 5-Step Template

Step 1. Set up. Choose recursive vs. sequential. List state, controls, constraints.
Step 2. FOCs + Envelope ⇒ Euler. Take FOC w.r.t. each control. Apply Envelope to each state. Substitute Envelope into FOC to eliminate V'.
Step 3. Define equilibrium. List {prices, allocations}, optimization, market clearing, consistency.
Step 4. Specialize for closed form. Apply log + Cobb–Douglas + $\delta = 1$ + iid as available. Use guess-and-verify if natural.
Step 5. Comparative statics or claim evaluation. Identify the GE channels: prices respond, market clears, distributions shift. Don't argue from the partial-equilibrium intuition alone.

The remainder of Part II walks through the mechanics of each step.

2.3 First-Five-Minutes Protocol

When you receive the exam:

1. **Read all three problems.** Identify their categories using the trigger table. Spend ≤ 3 minutes total.

2. **Allocate time.** The exam typically has three problems; budget ∼1 hour each. If one problem is in your weaker area, give it slightly more, but never less than 45 minutes.

3. **Within each problem, scan all subparts (a)–(e) before writing.** The earlier subparts often set up notation and assumptions used in later subparts. The closed-form simplification (log, CD, $\delta = 1$) often appears only in subpart (c) or (d) — knowing this lets you keep things general earlier.

4. **Identify the "Economist X" subpart.** If present, flag it. It is high-leverage: short to write, often the difference between a pass and a fail.

5. **Identify the equilibrium-definition subpart.** Always present somewhere; budget time to write it precisely (Sections 3.3.1–3.3.3).

2.4 What State Variables Should I Use?

Misidentifying the state is the most common Step-1 error. The reliable rule: **the state is everything that varies across the agent's decision problem and that the agent cannot perfectly forecast.**

- In a planner problem with capital, K alone (plus any exogenous shock).
- In an asset-pricing problem, the individual state is the agent's holdings a (stocks, bonds); the aggregate state is the dividend z (and any other exogenous source of price variation, e.g. time-varying net supply $\bar{y}$ in 2018 Q1, the tax rate τ in 2021 Q2 if it varies).
- In a heterogeneous-agent RCE, the individual state is a plus any idiosyncratic component (employment status, skill draw z); the aggregate state is the cross-sectional distribution of a *plus* any aggregate shock.
- In a search model with assets, the individual state is $(a, \text{employed/unemployed})$ for a worker who has not yet drawn this period's wage. After drawing, w enters the state.
- In a problem with an irreversible decision (e.g. 2019 Q3 employment, 2022 Q1 project), the state is $(a, \text{decision indicator})$ if past the decision, a alone if before.

Pitfall: Forgetting an Aggregate State

In a recursive competitive equilibrium with aggregate shocks (e.g. exogenous government spending G in 2024 Q3, or stochastic dividend z in any Lucas-tree problem), the aggregate state must enter both individual value functions *and* the equilibrium price functions. A common error is to write $p(z)$ but $V(a)$ instead of $V(a, z)$.

3 Part II — Standard Operating Procedures

This part walks through the mechanical moves you make on essentially every problem. The goal is to internalize them so you can execute steps 1–3 of the structural template on autopilot, freeing cognitive bandwidth for the hard subparts.

3.1 Setting Up the Bellman Equation: Three Templates

3.1.1 Template A: Single-Agent Decision Problem (Deterministic)

Used for planner problems with one representative type, or single-agent decision problems without shocks.

State: k (capital, asset, or skill stock). Choice: k' (next-period state) and possibly other controls (h hours, i investment, etc.).

$$V(k) = \max_{k',\text{other}} \left\{u(c) + \beta V(k')\right\}$$

subject to feasibility: c + investment + (anything saved) $\leq$ resources(k, other controls).

Standard examples:

- Optimal growth (2016 Q1, 2020 Q1): $V(K) = \max_{K'} u(F(K,h) + (1-\delta)K - K') + \beta V(K')$ where h may also be a choice.
- Endogenous growth (2017 Q2): replace $F(K)$ with $\Phi K^{\theta}(AK)^{1-\theta}$ with A depending on aggregate K in equilibrium.
- Resource extraction (2018 Q3): $V(x) = \max_l u(f(l)) + \beta V((x-l)(1+g))$.
- Skill accumulation (2021 Q3): $V(s) = \max_i u(ws - i) + \beta V((1-\delta)s + As^{1-\gamma}i^{\gamma})$.

3.1.2 Template B: Single-Agent Stochastic Decision Problem

Used when there is an exogenous shock z (dividend, technology, idea) following Markov process $Q(z' \mid z)$.

State: (a, z). Choice: a' (and other controls).

$$V(a,z) = \max_{a'} \left\{u(c) + \beta \int V(a',z')\, Q(\mathrm{d}z' \mid z)\right\}$$

subject to budget constraint linking c, a, a', and prices that may depend on z.

Standard examples:

- Lucas tree (2016 Q3, 2017 Q1, 2018 Q1, 2019 Q1, 2021 Q2, 2023 Q2): $V(a,z) = \max_{a'} u(c) + \beta\mathbb{E}[V(a',z') \mid z]$ with $c + p(z)a' = (p(z)+d(z))a$.
- AK with disasters (2024 Q1): the "shock" is binary disaster realization; capital stock evolves $k' = \rho k$ with prob. q, $k' = Ak - c$ with prob. $1-q$ (if no disaster wipes it out before consumption).
- Project choice (2022 Q1): the recursive structure differs after the decision is made, since one branch (failure) collapses to a deterministic continuation.

3.1.3 Template C: OLG — Sequential, Not Recursive

OLG problems are usually formulated *sequentially*, not recursively. The young agent solves a finite-horizon problem (live two or three periods), so dynamic programming is unnecessary.

Two-period OLG canonical setup:

$$\begin{aligned} \max_{c_{1,t}, c_{2,t+1}, s_t} \quad & \ln c_{1,t} + \beta \ln c_{2,t+1} \\ \text{s.t.} \quad & c_{1,t} + s_t = w_t - \tau_t \\ & c_{2,t+1} = (1 + r_{t+1}) s_t + p_{t+1} \end{aligned}$$

Three-period OLG (2018 Q2): Lifetime budget constraint substitutes through:

$$c_{1,t} + \frac{c_{2,t+1}}{R_{t+1}} + \frac{c_{3,t+2}}{R_{t+1} R_{t+2}} \le \text{lifetime income.}$$

Then maximize the lifetime utility subject to this single constraint (Lagrangian on a single multiplier, much faster than period-by-period FOCs).

3.2 FOCs and Envelope: The Mechanics

3.2.1 The General Algorithm

For Templates A or B, the algorithm is:

1. Substitute the budget constraint into the objective so the only choice variable is the next-period state (k' or a'). This yields an unconstrained max.
2. Take FOC with respect to k' (or a'). This will involve $V'(k')$ on the RHS.
3. Apply the Envelope theorem: differentiate the value function w.r.t. the state k, holding the optimal choice fixed. The Envelope condition expresses $V'(k)$ in current-period objects only.
4. Roll the Envelope forward by one period: $V'(k')$ in terms of period-(t+1) marginal utilities and prices.
5. Substitute into the FOC. Result: the Euler equation, an equation in c_t, c_{t+1}, prices, and shocks — no V.

3.2.2 Worked Example: Standard Growth (Deterministic)

Bellman: $V(k) = \max_{k'} u(f(k) + (1-\delta)k - k') + \beta V(k')$.
Let $c = f(k) + (1-\delta)k - k'$.
FOC (k'): $-u'(c) + \beta V'(k') = 0$, i.e. $u'(c) = \beta V'(k')$.
Envelope (k): $V'(k) = u'(c) \cdot [f'(k) + 1 - \delta]$.
Roll forward: $V'(k') = u'(c')[f'(k') + 1 - \delta]$.
Substitute into FOC:

$$u'(c_t) = \beta u'(c_{t+1}) \big[f'(k_{t+1}) + 1 - \delta \big]. \tag{1}$$

This is the deterministic consumption Euler equation.

3.2.3 Worked Example: Lucas Tree (Stochastic)

Bellman: $V(a, z) = \max_{a'} u((p(z) + d(z))a - p(z)a') + \beta\mathbb{E}[V(a', z') \mid z]$.
FOC (a'): $-u'(c)p(z) + \beta\mathbb{E}[V_a(a', z') \mid z] = 0$.
Envelope (a): $V_a(a, z) = u'(c)(p(z) + d(z))$.
Substitute:

$$p(z)u'(c) = \beta\mathbb{E}\left[u'(c')(p(z') + d(z')) \,\middle|\, z\right]. \tag{2}$$

3.2.4 Two-Asset Case (Lucas with Bond, 2023 Q2)

When there are multiple assets, you get one Euler equation per asset; their RHSs share the SDF $\beta u'(c')/u'(c)$.

For stock with price $p^s(z)$: $\quad p^s(z) = \mathbb{E}\left[\frac{\beta u'(c')}{u'(c)}(p^s(z') + d(z')) \,\middle|\, z\right]$.

For risk-free bond with price q: $\quad q = \mathbb{E}\left[\frac{\beta u'(c')}{u'(c)} \,\middle|\, z\right]$.

The risk-free rate is $1/q$. The expected return on the stock minus $1/q$ is the equity premium.

Trick: Bond Price When iid Shocks

If shocks are iid *and* consumption is constant in equilibrium (e.g. a Lucas-tree economy where all dividends are consumed and the dividend has a degenerate marginal distribution conditional on equilibrium), the bond price is constant: $q = \beta\mathbb{E}[u'(c')/u'(c)] = \beta$ when consumption is constant. The risk-free rate equals $1/\beta$, the time discount rate.

3.3 Equilibrium Definitions: Three Templates

This is the highest-yield section per minute of study. Definitions are short, formulaic, and graded strictly.

3.3.1 Template 1: Recursive Competitive Equilibrium (RCE)

RCE Template

A recursive competitive equilibrium consists of:

1. A set of value functions $V(\cdot)$ (one per type if heterogeneous) defined on the individual + aggregate state.
2. A set of decision rules / policy functions $g(\cdot)$ (consumption, savings, hours, labor supply, etc.).
3. A set of price functions $p(\cdot)$ (asset prices, wage, interest rate) defined on the aggregate state.
4. An aggregate law of motion Γ for the aggregate state.

such that:

(i) Given prices, value functions V and policy functions g solve the individual's Bellman equation.
(ii) Markets clear: e.g. asset market $\sum a' =$ supply; labor market $\sum h = \sum n^d$; goods market (often follows by Walras).
(iii) The aggregate law of motion Γ is consistent with individual decisions: aggregating g across agents reproduces Γ.

What graders look for: (a) clean separation of individual state from aggregate state; (b) explicit listing of *all* equilibrium objects; (c) the consistency condition (item iii). Forgetting consistency is the most common deduction.

Heterogeneous-agent variants (e.g. 2024 Q3 entrepreneur/worker, 2020 Q2 employed/unemployed): the aggregate state typically includes the cross-sectional distribution μ. Decision rules and value functions are indexed by type or status (worker W, entrepreneur E; employed e, unemployed u).

3.3.2 Template 2: Competitive Equilibrium with a Sequence of Markets

Used in OLG and growth contexts where you don't want recursive language.

Sequential Competitive Equilibrium Template

Allocations $\{c_t, k_t, h_t\}_{t=0}^{\infty}$ and prices $\{w_t, r_t\}_{t=0}^{\infty}$ such that:

(i) Given $\{w_t, r_t\}$, allocations solve each agent's optimization problem.
(ii) Firms maximize profits taking prices as given (typically: $w_t = F_L$, $r_t = F_K - \delta$).
(iii) All markets clear in every period.
(iv) (For OLG) Initial conditions (K_0, old generation endowment) are consistent.

3.3.3 Template 3: Arrow–Debreu Equilibrium

Used when prices are dated at $t = 0$ and there is one round of trading.

Arrow–Debreu Template

Allocations $\{c_t, k_t, h_t\}_{t=0}^{\infty}$ and a price system $\{p_t, w_t, r_t\}_{t=0}^{\infty}$ (or, in the Ramsey setting, also $\{\tau_t^l, \tau_t^k\}$) such that:

(i) Households maximize lifetime utility subject to the single date-0 budget constraint:

$$\sum_t p_t c_t + \sum_t p_t k_{t+1} \leq (\text{initial wealth}) + \sum_t p_t \big[w_t h_t (1 - \tau_t^l) + (1 - \delta + r_t(1 - \tau_t^k))k_t\big].$$

(ii) Firms maximize profits each period.
(iii) Government budget constraint holds in present value (intertemporal).
(iv) Markets clear in every period.

When to use which template? Use the language the question asks for. If it just says "define the equilibrium," use the framework most natural to the setup — recursive if value functions appear, sequential if no shocks and OLG-style.

3.4 The Stochastic Euler Equation: Quick Sign-Check

The Lucas-tree Euler equation (2) is the most reused formula on the exam. Recognize it in its various forms:

Form	Equation
Stock	$1 = \mathbb{E}\left[\frac{\beta u'(c')}{u'(c)} \cdot \frac{p(z')+d(z')}{p(z)} \,\middle|\, z\right]$
Stock with tax (2021 Q2)	$p(z)u'(c) = \beta \mathbb{E}\Big[u'(c')(1-\tau)(p(z') + d(z')) \,\Big|\, z\Big]$
Bond	$q = \beta \mathbb{E}\left[\frac{u'(c')}{u'(c)} \,\middle|\, z\right]$
Two trees, no bond (2017 Q1)	$p_i(z)u'(c) = \beta \mathbb{E}\Big[u'(c')(p_i(z') + z_i') \,\Big|\, z\Big], \quad i = 1, 2$

Sign-check: the price today equals the expected discounted payoff tomorrow. The discount factor is the SDF $\beta u'(c')/u'(c)$. If you don't see this structure, recheck.

4 Part III — Closed-Form Recipes

When a problem says "Suppose $u(c) = \ln c$, the production function is Cobb–Douglas, and $\delta = 1$," it is asking you to recognize one of the recipes below. Each appears multiple times across the sample exams. Memorize the result *and* the trick, not just the formula.

4.1 The "Holy Trinity": Log + Cobb–Douglas + $\delta = 1$

This combination is the single most common closed-form trigger. It appears in: 2016 Q1 (planner growth, log only); 2018 Q2 (3-period OLG); 2019 Q2 (OLG with college); 2020 Q1 (growth with leisure); 2020 Q3 (OLG with social security); 2021 Q1 (OLG two-person households); 2022 Q2 (OLG home production); 2017 Q3 (OLG with infrastructure); among others.

Why does it work? With log utility, income and substitution effects on saving exactly cancel, so the saving rate is independent of the rate of return. With Cobb–Douglas, factor shares are constant. With $\delta = 1$, capital tomorrow equals investment today (no $(1-\delta)k$ term to track). The combination yields linear policies.

4.1.1 Recipe 1: Single-Agent Growth, Log + CD + $\delta = 1$

Recipe 1

Setup: $u(c) = \ln c$, $f(k) = k^\theta$, $\delta = 1$, no labor choice.
Bellman: $V(k) = \max_{k'} \ln(k^\theta - k') + \beta V(k')$.
Guess: $V(k) = A + B \ln k$ for constants A, B.
Verify: Substitute and match coefficients. The optimal policy is:

$$k_{t+1} = \beta\theta \cdot k_t^\theta, \qquad c_t = (1 - \beta\theta) \cdot k_t^\theta.$$

Saving rate: $\beta\theta$ (constant). **Steady state:** $k^* = (\beta\theta)^{1/(1-\theta)}$.

4.1.2 Recipe 2: Logarithmic OLG with Capital

Recipe 2

Setup: Two-period OLG, log utility, work only when young, wage w_t, return $R_{t+1} = 1 + r_{t+1}$ (with $\delta = 1$, R_{t+1} equals the gross return; otherwise $R_{t+1} = 1 - \delta + r_{t+1}$).
FOCs:

$$\begin{aligned} &\text{(young)} & \frac{1}{c_{1,t}} &= \lambda_t \\ &\text{(old)} & \frac{\beta}{c_{2,t+1}} &= \lambda_t / R_{t+1} \\ & \Rightarrow & c_{2,t+1} &= \beta R_{t+1}\, c_{1,t} \end{aligned}$$

Saving function: $s_t = \dfrac{\beta}{1+\beta}(w_t - \tau_t)$, where τ_t is any tax/contribution paid when young.

Key feature: Saving rate $\beta/(1+\beta)$ is independent of R_{t+1} (income/substitution cancellation under log).

Why this matters: The closed-form saving function plus the firm FOCs ($w_t = (1-\alpha)k_t^\alpha$ and the per-worker resource constraint $K_{t+1} = N_t s_t$) give the law of motion for k:

Law of motion (no growth, no taxes): $k_{t+1} = \dfrac{\beta(1-\alpha)}{(1+\beta)(1+n)} k_t^\alpha$, where n is population growth.

Steady state: $k^* = \left[\dfrac{\beta(1-\alpha)}{(1+\beta)(1+n)}\right]^{1/(1-\alpha)}$.

4.1.3 Recipe 3: OLG with Social Security (2020 Q3)

Recipe 3

Setup: Recipe 2, plus $d_t = \tau w_t$, balanced budget pension p_{t+1}.
Government BC: $N_t^o p_t = N_t^y \tau w_t \Rightarrow p_t = (1+n)\tau w_t$ (using $N_t^y = (1+n)N_t^o$).
Effective lifetime budget:

$$c_{1,t} + \frac{c_{2,t+1}}{R_{t+1}} = (1-\tau)w_t + \frac{(1+n)\tau w_{t+1}}{R_{t+1}}$$

Saving: $s_t = \dfrac{\beta}{1+\beta}(1-\tau)w_t - \dfrac{1}{1+\beta}\cdot\dfrac{(1+n)\tau w_{t+1}}{R_{t+1}}$. Saving falls in τ, both directly (less disposable income when young) and indirectly (the implicit pension wealth substitutes for private saving).
Steady-state k^* falls in τ.

4.2 The Lucas Tree with Log Utility

Recipe 4: Log Lucas Tree

Setup: Single tree, dividends z_t Markov; representative agent with $u(c) = \ln c$; equilibrium $c_t = z_t$ (market clearing).
Euler: $p(z)/z = \beta\mathbb{E}[(p(z') + z')/z' \mid z]$.
Trick: Conjecture $p(z) = \phi z$ for constant ϕ. Substitute:

$$\phi = \beta\mathbb{E}\left[\frac{\phi z' + z'}{z'}\,\middle|\, z\right] = \beta(\phi+1)$$

Result: $\phi = \dfrac{\beta}{1-\beta}$, so $\boxed{p(z) = \dfrac{\beta}{1-\beta}z}$.

This is robust across many problems:

- **Two trees iid (2016 Q3):** prices are linear in their own dividend; same conjecture works for each tree.
- **Two trees with sum=1 (2017 Q1):** consumption equals $z_1 + z_2 = 1$, constant! So $u'(c)$ is constant; prices reduce to $p_i(z) = \beta\mathbb{E}[p_i(z') + z_i' \mid z]$ — a linear functional equation that solves to give $p_1 + p_2 = \beta/(1-\beta)$ (independent of z).
- **With taxation (2021 Q2):** the after-tax payoff $(1-\tau)(p+d)$ replaces $(p+d)$. Conjecture $p = \phi z$ still works, with $\phi = \beta(1-\tau)/(1-\beta(1-\tau))$.

- **With bond in utility (2023 Q2):** consumption is still z in equilibrium, but the marginal utility is $u'(z+\kappa b)$ which evaluated at equilibrium $b=0$ gives $u'(z)$. The bond price is the standard SDF expectation, but the equity premium captures Jensen-style risk effects.

4.3 AK Growth with Log Utility

Recipe 5: Log + AK

Setup: $f(k)=Ak$, full depreciation, no labor. $V(k)=\max_{k'}\ln(Ak-k')+\beta V(k')$.
Guess: $V(k)=a+b\ln k$. Substitute and solve.
Result: $k_{t+1}=\beta A k_t$, $c_t=(1-\beta)Ak_t$.
Aggregate stock grows at rate βA per period.

Where it appears:

- 2024 Q1 (AK with disasters): the disaster is binary; modify the Bellman to include the disaster probability. With CRRA, the saving rate becomes $\beta^{1/\sigma}[\rho q+(1-q)A]^{(1-\sigma)/\sigma}/A$ multiplied appropriately — see Section 4.4.
- 2023 Q3 (alternating management): the aggregate analog with linear individual policies.
- 2017 Q2 (endogenous-growth externality): A is endogenous; the equilibrium has a growth rate that differs from the planner's.

4.4 CRRA Growth with Linear Technology (Disaster Problem, 2024 Q1)

Recipe 6: CRRA + AK + Disaster Risk

Setup: $u(c)=(c^{1-\sigma}-1)/(1-\sigma)$, $f(k)=Ak$, capital wiped to ρk with prob. q.
Bellman: $V(k)=\max_{c,k'}u(c)+\beta\big[(1-q)V(k')+qV(\rho k')\big]$, with $c+k'=Ak$ (when investing today, the disaster strikes *tomorrow*, so today's k' becomes $\rho k'$ next period if disaster hits).
Guess: $c=\phi Ak$ (constant saving rate). Verify via Euler:

$$u'(c)=\beta\big[(1-q)Au'(c')+q\rho Au'(c'')\big]$$

where c' is no-disaster consumption next period and c'' is post-disaster consumption.
Result:

$$\boxed{1-\phi=(\beta A)^{1/\sigma}\big[(1-q)+q\rho^{1-\sigma}\big]^{1/\sigma}/A}$$

(rearranged version of the Euler identity).
Comparative statics:

- Higher q: ambiguous. Two effects — precautionary saving (raises ϕ, the saving rate) and lower expected return (lowers ϕ). Direction depends on whether $\sigma \gtrless 1$.
- Lower ρ (more severe disasters): same trade-off, sign depends on $\sigma \gtrless 1$.

The intuition for $\sigma \gtrless 1$:

- $\sigma>1$: agents are very risk averse; the income effect dominates and they save *more* when disaster risk rises (precautionary motive).
- $\sigma<1$: substitution effect dominates; agents save *less* because the expected return on saving is lower.

- $\sigma = 1$ (log): exact cancellation — saving rate independent of q, ρ.

4.5 McCall Reservation Wage (Closed Forms)

Recipe 7: McCall, No Assets and No Quitting

Setup: Risk-neutral worker (no consumption smoothing); unemployment benefit b; wage offers $w \sim F$ iid.

Reservation wage equation:

$$w^* = b + \frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} (1 - F(w))\, \mathrm{d}w$$

(integration by parts of the standard equation).

Comparative statics:

$$\frac{\partial w^*}{\partial b} > 0, \quad \frac{\partial w^*}{\partial \beta} > 0, \quad \text{mean-preserving spread of } F \Rightarrow w^* \uparrow .$$

4.5.1 Search with Assets (2020 Q2)

When the worker has access to a bond market, the reservation wage depends on a. The key result:

Result: Reservation Wage Increasing in Assets

If u is strictly concave and the employed-vs-unemployed value functions satisfy $V_e(a, w^*) = V_u(a)$ at the cutoff, then *if* consumption when employed exceeds consumption when unemployed at the same a, the reservation wage $w^*(a)$ is strictly increasing in a.

Intuition: Higher a allows the unemployed to consume more this period (higher V_u), making her pickier about offers. *This is the essence of the proof in 2020 Q2.*

4.5.2 Search with Temporary UE Compensation (2022 Q3)

The reservation wages satisfy $w_e^* < w_+^* < w_1^*$ (note ordering): an employed worker is most willing to keep her job; a worker still eligible for benefits is pickiest.

Intuition for $w_1^* > w_+^*$: The eligible worker can "keep waiting" next period at cost only $-b$ relative to no benefit; the option value is higher. Hence she demands higher wage to accept a job and forgo the benefit.

Difference formula (2022 Q3d):

$$w_1^* - w_+^* = b - \beta\, b \cdot (\text{prob. of needing it})$$

which simplifies to roughly b in the worker's units.

4.6 Quick Reference: Closed-Form Card

Setup	**Result**	**Recipe**
Log + CD + $\delta = 1$ growth	$c = (1-\beta\theta)k^{\theta}$, save rate $\beta\theta$	1
Log OLG + CD + $\delta = 1$	$s = \frac{\beta}{1+\beta}w$, $k^* = \left[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)}\right]^{1/(1-\alpha)}$	2
OLG + Social Security	s falls in τ, k^* falls in τ	3
Log Lucas tree	$p(z) = \frac{\beta}{1-\beta}z$	4
Log AK growth	$k' = \beta Ak$, $c = (1-\beta)Ak$	5
CRRA + AK + disaster	saving rate ϕ from Euler; sign of $\partial\phi/\partial q$ depends on σ	6
McCall reservation wage	$w^* = b + \frac{\beta}{1-\beta}\int_{w^*}^{\bar{w}}(1-F)$	7
Two trees, sum constant	$p_1 + p_2 = \beta/(1-\beta)$ (Section 4.1)	4 (var.)
Lucas with proportional tax τ	$p(z) = \frac{\beta(1-\tau)}{1-\beta(1-\tau)}z$	4 (var.)

5 Part IV — Proof Techniques

The exam frequently asks "Show that..." or "Prove that..." These are not closed-form computations; they require an argument. Below is a catalog of the recurring proof patterns and how to execute each.

5.1 Cutoff / Threshold Proofs

Pattern. The problem has a discrete choice (project S vs. R, work vs. not work, eligible vs. not). You must show there exists a threshold value of some state variable above (or below) which the agent prefers one option.

Standard structure of argument:

1. Let $V_A(x)$ and $V_B(x)$ be the value functions under each option, as a function of the state x.
2. Show $V_A(x) - V_B(x)$ is strictly monotonic in x (use envelope/derivative or single-crossing).
3. Show $V_A - V_B$ takes both signs over the relevant range (limit arguments).
4. Conclude: by intermediate value theorem and monotonicity, there is a unique cutoff $\bar{x}$ at which $V_A(\bar{x}) = V_B(\bar{x})$.

5.1.1 Worked Example: 2022 Q1 (Project Choice)

Claim: There exists $\bar{k}_0 \geq 0$ such that the agent picks the safe project S iff $k_0 \geq \bar{k}_0$.

Sketch:

1. Let $V_S(k)$ and $V_R(k)$ be the lifetime values of operating projects S and R starting with capital k. Both involve a deterministic part (the surviving project's growth path) and a constant continuation value $W = \sum_t \beta^t \ln b = \ln b/(1-\beta)$ (when the project fails).
2. For each project, the agent solves a standard log + AK problem until failure. The solution is linear in k, and lifetime utility from operating project i is $V_i(k_0) = \alpha_i \ln k_0 + \beta_i$ for some constants α_i, β_i determined by A_i, γ_i, β.
3. Show that as $k_0 \to 0$, the agent prefers R (the higher continuation value W from failing dominates — benefit b is more valuable when k_0 is small). As $k_0 \to \infty$, the agent prefers S (the higher survival probability matters more when there is more to lose).
4. Monotonicity in k_0 of the difference $V_S(k_0) - V_R(k_0)$ ensures uniqueness of the cutoff.

Intuition to give in the answer: the safe project is preferred when the stakes are larger because the survival probability matters more in absolute terms; the risky project is preferred when stakes are small because the consumption-insurance benefit b is more attractive relative to small-scale operations.

5.1.2 Worked Example: 2019 Q3 (Irreversible Employment)

Claim: The agent's employment choice is characterized by a threshold $\bar{a}_0$ such that the agent works iff $a_0 \leq \bar{a}_0$ (the wealthy don't bother working).

Sketch:

1. Derive $V_1(a_0)$ (work value) and $V_0(a_0)$ (don't-work value) by solving the consumption Euler equation along each path.

2. Both are strictly increasing in a_0. Argue that $V_1 - V_0$ is strictly decreasing in a_0: the marginal benefit of the labor income y decreases (decreasing marginal utility of consumption); the marginal cost ϕ stays constant.
3. At $a_0 = 0$: working is strictly preferred (without working, no income at all, $V_0 \to -\infty$).
4. As $a_0 \to \infty$: not working is strictly preferred (high consumption from initial wealth makes the disutility ϕ relatively more important).
5. Conclude: unique cutoff $\bar{a}_0$.

5.2 Single-Crossing Argument for Reservation Wages

Pattern. In a search model, the worker's optimal strategy is to accept iff $w \geq w^*$, where w^* is determined by indifference. To prove the reservation-wage form, exhibit a single-crossing of $V_e(w)$ and V_u in w.

Argument template:

- $V_e(w)$ is strictly increasing in w (employment with higher w dominates lower w).
- V_u does not depend on w (a rejected offer is rejected; future offers are iid).
- Therefore the difference $V_e(w) - V_u$ is strictly increasing in w, crosses zero at most once. The crossing point is w^*.

5.2.1 Worked Example: 2020 Q2 (Search with Assets)

Claim: If $c_e(a) > c_u(a)$, then $w^*(a)$ is strictly increasing in a.

Sketch:

1. Cutoff condition: $V_u(a) = V_e(a, w^*(a))$, i.e. at the reservation wage the worker is indifferent.
2. Differentiate both sides w.r.t. a using the Envelope theorem applied to each value function:
$$V_u'(a) = u'(c_u(a))(1+r), \qquad V_{e,a}(a, w) = u'(c_e(a))(1+r).$$
3. Total derivative of the indifference condition: $V_u'(a) = V_{e,a}(a, w^*(a)) + V_{e,w}(a, w^*(a)) \cdot \frac{\partial w^*}{\partial a}$.
4. Solve: $\dfrac{\partial w^*}{\partial a} = \dfrac{V_u'(a) - V_{e,a}(a, w^*(a))}{V_{e,w}(a, w^*(a))} = \dfrac{(1+r)[u'(c_u) - u'(c_e)]}{V_{e,w}}$.
5. Numerator: $u'(c_u) > u'(c_e)$ since $c_e > c_u$ and u is strictly concave. Denominator: $V_{e,w} > 0$. Hence $\partial w^* / \partial a > 0$. **Q.E.D.**

5.2.2 Worked Example: 2022 Q3 (Reservation Wage Ordering)

Claim: $w_e^* < w_+^* < w_1^*$.

Sketch: Compare the indifference conditions. The employed worker (just before the period starts) loses an inframarginal stream of w if she quits. Forfeiting this stream raises her reservation wage to keep the job (so w_e^* is the lowest among the three). The unemployed-and-eligible worker has the highest outside option (the benefit b this period), so she is the choosiest. Formal argument: write the three indifference conditions and apply algebra. The differences are:

$$w_1^* - w_+^* = b\,(1 - \beta \cdot \text{Prob[next period no benefit]})$$

Implication for the qualitative claim: workers with longer past unemployment have lower reservation wages, so they accept a wider range of jobs. Conversely, workers most recently unemployed are the choosiest. *The "more likely to quit" implication*: a worker entering employment from a long unemployment spell has lower reservation wage w, hence is closer to the quit threshold w_e^* than someone with a short unemployment spell.

5.3 Guess-and-Verify

Pattern. For log utility with multiplicative production (CD, AK, or skill) and full depreciation, conjecture a linear policy $g(k) = \phi k$ for some unknown ϕ. Substitute into the Bellman, solve for ϕ.

5.3.1 Worked Example: 2021 Q3 (Skill Accumulation)

Setup: $s_{t+1} = (1-\delta)s_t + A s_t^{1-\gamma} i_t^{\gamma}$, $u(c) = \ln c$, $c_t = w s_t - i_t$.

Guess: $V(s) = \phi_0 + \phi_1 \ln s$ for constants. Then policy $i(s) = \xi s$ is linear in s.

Verify:

1. FOC w.r.t. i: $u'(c) = \beta V'(s') \cdot \gamma A s^{1-\gamma} i^{\gamma-1}$.
2. Envelope: $V'(s) = u'(c) w + \beta V'(s') \cdot [(1-\delta) + (1-\gamma) A s^{-\gamma} i^{\gamma}]$.
3. Substitute the conjectured V and solve for ξ.
4. The resulting ξ depends on $A, \gamma, \beta, \delta, w$.

Effect of δ: Implicit differentiation gives $\partial\xi/\partial\delta < 0$. Higher depreciation means the next period's skill stock is more eroded; this reduces the marginal benefit of investing today (the asset depreciates more rapidly), so the optimal investment falls.

5.4 Monotone Comparative Statics on Equilibrium Objects

Pattern. "Show that the equilibrium price/quantity is increasing/decreasing in parameter θ."

Standard approach:

1. Write the equilibrium condition (often the FOC, market clearing, or steady-state equation) implicitly: $H(p, \theta) = 0$.
2. Differentiate: $\partial p/\partial\theta = -H_\theta / H_p$.
3. Sign each partial derivative using the structure of H.

5.4.1 Worked Example: 2021 Q2 (Asset Prices Decreasing in Tax Rate)

Claim: Equilibrium asset prices in all states are strictly decreasing in τ.

Sketch using the contraction mapping:

1. Equilibrium price function $p(z; \tau)$ satisfies the operator equation $T_\tau p = p$, where

$$(T_\tau p)(z) = (1-\tau)\beta \int \frac{u'(z')}{u'(z)} (p(z') + z')\, Q(\,\mathrm{d}z'|z)$$

   (using $c = z$ in equilibrium and a stylized lump-sum rebate that doesn't enter the no-arbitrage condition).

2. T_τ is a contraction; let $p^*(z;\tau)$ be the unique fixed point.
3. Show T_τ is monotone in τ: if $p > p'$ pointwise, then $T_\tau p > T_\tau p'$ pointwise. Also τ enters as a multiplicative factor $(1-\tau)$ that is decreasing in τ.
4. Use Blackwell-type comparison: a contraction with a parameter that monotonically shifts the operator monotonically shifts the fixed point.
5. Conclude $p^*(z;\tau)$ is strictly decreasing in τ.

Intuition: the after-tax payoff per unit asset is reduced; agents are willing to pay less to hold the asset. Pure level effect; rebate is lump-sum and doesn't undo the wedge on intertemporal choice.

5.5 Existence / Uniqueness of Steady State

Pattern. "Show that there exists a unique steady-state capital-to-labor ratio."

Standard approach in OLG with log + CD:

1. Write the law of motion $k_{t+1} = \Psi(k_t)$.
2. Steady state: k^* solves $k^* = \Psi(k^*)$.
3. In log + CD, $\Psi(k) = \phi k^\alpha$ for some $\phi > 0$, $\alpha \in (0,1)$. The function $\Psi(k) - k$ is positive for k small (Inada-style) and negative for k large (concavity), and continuous, monotonic in the relevant range. By IVT, there exists a positive k^*. Uniqueness follows from strict concavity of Ψ.

Two traps to avoid:

- Don't forget the trivial steady state $k^* = 0$. Argue separately that the positive steady state is the relevant one (Inada conditions, or the dynamics push away from zero).
- In two-type OLG (heterogeneous agents), the law of motion uses the population-weighted aggregate savings, which is still of the form ϕk^α with ϕ depending on type composition.

5.6 Inequality of Consumption Across States

Pattern. "Prove that $c_1(a) > c_0(a)$ for any a."

Standard approach: The two consumption functions arise from solving two different optimization problems with different income paths. The agent with higher (permanent) income consumes more at each a *provided* the assets-feasibility constraints don't bind in the same way.

5.6.1 Worked Example: 2019 Q3

Sketch:

1. Both agents face the same Euler equation $u'(c_t) = \beta(1+r)u'(c_{t+1})$.
2. Working agent's lifetime budget includes income $y/(1-(1+r)^{-1}\cdot)$ in present value, which is strictly larger than the non-working agent's.
3. Higher lifetime resources $\Rightarrow$ higher consumption at every period $\Rightarrow$ in particular at $t = 0$, given the same a_0.
4. Working agent's consumption path is uniformly higher (parallel level shift in CRRA utility, or proportional shift in log).

Care: the no-borrowing constraint $a_t \geq 0$ may bind for the non-working agent earlier. The result depends on the working agent never being borrowing-constrained, which is the assumption from part (b) of the problem.

5.7 The 2023 Q3 Trick: Linear Saving Rule with Strategic Interaction

This is the most novel proof of the recent set, worth highlighting.

Setup: Two agents jointly own an AK asset. Alternating management. Each agent's value depends on the *other* agent's saving rule.

Trick: Conjecture that both agents use linear saving rules: $g_i(k) = \mu_i k$ for $i = 1, 2$. By symmetry, $\mu_1 = \mu_2 = \mu$.

Verify:

1. Take agent 1's problem given agent 2 uses $g_2(k') = \mu k'$.
2. Substitute: $V_1(k) = \max_{c,c',k'} \ln c + \beta \ln c' + \beta^2 V_1(\mu k')$.
3. Guess: $V_1(k) = a + b \ln k$. Solve.
4. By symmetry, the same μ satisfies agent 2's problem.

Compared with independent management: The aggregate stock evolves $K' = 2\mu k$ (joint) versus $K' = 2\beta A k$ (independent). The joint case grows slower, because each agent free-rides on the other (and depletes the common pool more rapidly).

6 Part V — Topic Frameworks

This part gives a compact framework for each major topic. It is *not* a substitute for lecture notes; it is a triage reference. Each subsection: canonical setup, variations seen, common pitfalls.

6.1 Overlapping Generations (OLG)

6.1.1 Canonical 2-Period OLG with Capital

Time discrete, $t = 0, 1, 2, \ldots$. Each cohort lives 2 periods. Population grows at rate n. Agents work in period 1 (typically), supply 1 unit of labor inelastically, save s_t, consume in both periods. Cobb–Douglas production, capital fully depreciates.

Canonical equations:

- Saving: $s_t = \frac{\beta}{1+\beta}(w_t - \tau_t)$ when log.
- Wage: $w_t = (1-\alpha)k_t^\alpha$.
- Return: $R_{t+1} = \alpha k_{t+1}^{\alpha-1}$ (gross, since $\delta = 1$).
- Capital market clearing: $K_{t+1} = N_t^y s_t \Rightarrow k_{t+1} = s_t/(1+n)$.
- Law of motion: $k_{t+1} = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)} k_t^\alpha$.

6.1.2 Variations Table

Variation	What changes	Past exam
3-period OLG	Lifetime budget through 3 periods; mortality δ	2018 Q2
Heterogeneous skill	π skilled, $1-\pi$ unskilled; aggregate labor uses CES or linear aggregation	2016 Q2, 2019 Q2
Heterogeneous taste	$\gamma_L < \gamma_H$ valuation of home goods (or other)	2022 Q2
Two-person households	Joint consumption, joint disutility of work; cutoff in q	2021 Q1
With home production	Add $h = G(l)$ or Ah; second consumption good	2022 Q2, 2024 Q2
With social security	$d_t = \tau w_t$, balanced budget pension p_t	2020 Q3
With public infrastructure	z^λ in production; tax-financed	2017 Q3
Without capital	Pure consumption-loan economy; type-A vs type-B endowments	2023 Q1
Two periods, work both	Type C with growth, type NC flat	2019 Q2
With growth g	TFP grows at $(1+g)$; balanced growth path	2018 Q2, 2024 Q2
With age cohorts	Working-age vs retired; pension	2020 Q3

6.1.3 Common OLG Pitfalls

OLG Pitfalls

- **Wrong cohort dating.** An agent born at t is young at t, old at $t+1$. The capital they save at t enters production at $t+1$. Always write $K_{t+1} = N_t^y s_t$, not $K_t = N_t^y s_t$.
- **Population vs. labor.** If only the young work, $L_t = N_t^y$. If both young and old work (e.g. 2019 Q2), $L_t = N_t^y \cdot e_y + N_t^o \cdot e_o$ with appropriate efficiencies.
- **Wrong return.** With $\delta = 1$, the gross return $R = \alpha k^{\alpha-1}$. With $\delta < 1$, $R = 1 - \delta + \alpha k^{\alpha-1}$.
- **Forgetting initial old generation.** The OLG economy starts at $t = 0$ with an initial old cohort; their consumption is determined separately (typically equal to their savings plus return on initial K_0).
- **Pension formula.** If $N_t^y = (1+n)N_{t-1}^y$, then in a pay-as-you-go system $p_t \cdot N_{t-1}^y = \tau w_t \cdot N_t^y$, so $p_t = (1+n)\tau w_t$. The factor $(1+n)$ matters; missing it loses points.
- **Heterogeneity breaks aggregation.** If types have different saving rates, K_{t+1} = sum across types weighted by population shares. With log utility, save rate $\beta/(1+\beta)$ is universal across types, so aggregation is clean. With non-log, much messier.

6.2 Asset Pricing (Lucas Tree and Variants)

6.2.1 Canonical Single Tree

Mass 1 of agents, each maximizes $\mathbb{E}_0 \sum \beta^t u(c_t)$. Single tree paying dividends z_t following Markov Q. Agents trade tree shares; total supply = 1. In equilibrium, all hold the tree, $c_t = z_t$.

Pricing equation: $p(z)u'(z) = \beta \mathbb{E}[u'(z')(p(z') + z')|z]$.

6.2.2 Variations Table

Variation	What changes	Past exam
Two trees, iid	Two pricing equations; consumption = sum of dividends	2016 Q3
Two trees, sum constant	c constant in equilibrium; $u'(c)$ pulls out; sum of prices is constant	2017 Q1
Time-varying net supply	Aggregate state includes $\bar{y}_t$; alternating between $\bar{y} = 1$ and $\bar{y} = 2$	2018 Q1
Two countries + bond	Country-specific dividends; international bond market; segmented stock markets	2019 Q1
With proportional tax	After-tax payoff $(1-\tau)(p+d)$; lump-sum rebates don't enter pricing	2021 Q2
Bond in utility	$u(c + \kappa b)$; bond has direct consumption-equivalent value	2023 Q2

6.2.3 Asset Pricing Pitfalls

Asset Pricing Pitfalls

- **Forgetting market clearing.** When deriving the equilibrium price, you must use $c = z$ (or $c = z_1 + z_2$, etc.). Otherwise you have an arbitrary price function.
- **Mismeasuring the dividend.** In 2018 Q1 with time-varying net supply, the dividend per share z is constant, but the per-capita consumption is $c_t = \bar{y}_t z$, so consumption *does* fluctuate even though the dividend is constant. The aggregate state is $\bar{y}$, not z.
- **Wrong handling of taxes.** The tax in 2021 Q2 is on *income* = dividends + asset value. The lump-sum rebate is a transfer; it does not enter the no-arbitrage condition (Walras's law / Ricardian equivalence within the period).
- **Equilibrium vs. off-equilibrium.** The Euler equation evaluated at equilibrium uses $c = z$; the agent's individual Euler equation (before market clearing) uses her own c. Be explicit about which version you are presenting.

6.3 Search (McCall and Variants)

6.3.1 Canonical McCall Model

Risk-neutral worker. Each period, unemployed worker draws wage offer $w \sim F$ iid. Accept $\Rightarrow$ work at w forever (or until separation). Reject $\Rightarrow$ get b (unemployment benefit), draw again next period.

Bellman equations:

$$v_e(w) = w + \beta v_e(w) = \frac{w}{1-\beta}$$
$$v_u = b + \beta \int \max\{v_e(w'), v_u\}\, \mathrm{d}F(w')$$

Reservation wage: w^* such that $v_e(w^*) = v_u$, i.e. $\frac{w^*}{1-\beta} = b + \beta v_u$.

6.3.2 Variations

Variation	What changes	Past exam
With job separation ρ	Employed workers face exogenous probability ρ of becoming unemployed	2020 Q2
With assets	Concave u, save/borrow with bond, asset a in state	2020 Q2
With temporary UE compensation	Three reservation wages depending on UE history	2022 Q3
Heterogeneous workers	Different b, F, or skill — not seen in samples but possible	—

6.3.3 Search Pitfalls

Search Pitfalls

- **Timing of acceptance.** The convention used in the sample exams: an offer accepted today means the worker starts at the job *next* period. Be explicit about this.
- **Unemployment compensation eligibility.** In 2022 Q3, the agent must work at least one period to requalify for benefits. Tracking the eligibility status as a state variable is essential.
- **Wrong reservation wage equation.** The standard formula uses $v_e(w^*) = v_u$. With temporary UE comp. as in 2022 Q3, the comparison includes the current-period benefit b on the unemployment side. Use the correct v_1, v_+ distinction.
- **Concavity matters.** In 2020 Q2 with assets, you cannot use the simple closed-form reservation wage equation; the value functions depend on a and you must work with implicit functions.

6.4 Endogenous Growth and Ramsey Taxation

6.4.1 Endogenous Growth (2017 Q2)

Setup: $y_{i,t} = \Phi k_{i,t}^{\theta}(A_t l_{i,t})^{1-\theta}$ with $A_t = \int k_{i,t}\, \mathrm{d}i$ (externality).

In equilibrium (symmetric): $A_t = K_t = k_t$. So effective production is $y_t = \Phi k_t^{\theta}(k_t l_t)^{1-\theta} = \Phi k_t l_t^{1-\theta}$. With $l_t = 1$, this is AK with $A = \Phi$.

Equilibrium growth rate (consumption Euler with log + AK + $\delta = 1$): $c_{t+1}/c_t = \beta\Phi(1-\theta)$ (since the private return ignores the externality and equals the marginal product of own capital, $\Phi(1-\theta)$ in symmetric equilibrium with $l = 1$ — not Φ).

Wait — careful. Private return to capital: $\partial y/\partial k_i$ taking $A_t = K_t$ as given:

$$\partial y_i/\partial k_i = \theta\Phi k_i^{\theta-1}(A_t l_i)^{1-\theta} = \theta\Phi$$

in symmetric equilibrium with $k_i = A_t = k_t$ and $l_i = 1$.

Planner's return: accounts for the externality. Total derivative w.r.t. k_i when $A_t = k_i$ also moves: $(\theta + (1-\theta))\Phi = \Phi$.

Result:

- Equilibrium growth: $g^{eq} = \beta\theta\Phi$ (recipe 5 with $A = \theta\Phi$).
- Planner growth: $g^* = \beta\Phi$.
- Wedge: $\theta < 1$ implies $g^{eq} < g^*$.

Negative growth condition: $g^{eq} < 1 \Leftrightarrow \beta\theta\Phi < 1$. So for intermediate Φ (small enough that $\beta\theta\Phi < 1$ but large enough that $\beta\Phi > 1$), equilibrium growth is negative while planner's is positive.

Optimal policy: a labor income subsidy combined with a tax on labor that implements the planner's allocation. The wedge is $1/\theta$, so the implicit subsidy on capital is $(1-\theta)/\theta$. (Equivalently, set the labor income tax to fund a capital subsidy that brings the private return up to the social return.)

6.4.2 Ramsey Taxation (2016 Q4, 2017 Q4)

Standard moves:

1. Write the household's first-order conditions.

2. Derive the implementability constraint by substituting the FOCs and prices into the household budget constraint.

3. Substitute the resource constraint into the utility (replace c_t).

4. The Ramsey planner maximizes the resulting expression subject to the implementability constraint.

5. **Key result:** optimal capital tax is zero from period 1 onwards (unless there is a constraint on $\tau_0^k = 0$; otherwise it would be infinite in period 0).

Why zero capital tax? Capital taxation distorts intertemporal margins; a small capital tax compounds into a large distortion. It is optimal to use lump-sum-equivalent labor taxation instead.

For the 2017 Q4 specific case (no capital): the question is balanced budget vs. constant tax rates. Under quasi-linear preferences (linear in c, isoelastic in l), the equation $c_t - \frac{1}{1+\varphi} l_t^{1+\varphi}$ implies that the labor wedge is the same each period, but the deadweight loss from a tax is convex in the tax rate. Hence by Jensen's inequality, smoothing the tax rate (constant τ_t) is welfare-superior to lumpy taxation (balanced budget).

6.5 Decision Problems with Twists

This catch-all category includes Galina's signature problems with custom state spaces or strategic structure.

Common features:

- Often a non-standard law of motion (resource: $x' = (x - l)(1 + g)$; skill: $s' = (1 - \delta)s + As^{1-\gamma} i^{\gamma}$).
- Often an irreversible decision (project: 2022 Q1; employment: 2019 Q3).
- Often a closed-form available via guess-and-verify with log utility.
- Always require careful state specification.

Standard moves:

1. Identify state. Is it k alone? $(k, \text{decision indicator})$? (s_1, s_2) with two stocks?
2. Write the Bellman.
3. Look for a known special case: log + AK-like structure $\Rightarrow$ guess linear policy. CRRA + AK $\Rightarrow$ guess constant-saving-rate policy.
4. For irreversible decisions, characterize a cutoff.
5. For strategic problems (2023 Q3), conjecture symmetric linear policies.

7 Part VI — The "Economist X" Rubric

A subpart formatted as "Economist X claims Y — is she right? Explain" appears in at least 11 of the 29 sample problems. **The claim is almost always wrong, or right for the wrong reason.** The graders are testing whether you recognize the general-equilibrium feedback the claim ignores.

This section catalogues the recurring mistakes economists in past exams have made, along with the corrective argument you should give.

7.1 The Diagnostic Protocol

When you see this subpart:

1. **Restate the claim as a chain of implications.** "She is saying: *premise* $A \Rightarrow$ *conclusion* B via mechanism C."
2. **Identify which step the GE feedback breaks.** Usually it is the link from a partial-equilibrium intuition to the equilibrium statement.
3. **Either:** (i) state the missing channel and show it goes the other way, or (ii) confirm her conclusion but provide the correct mechanism (graders accept this — right answer for the wrong reason still passes if you supply the right reason).
4. **Be precise about the comparative statics.** Steady state vs. transition, partial vs. general, individual vs. aggregate.

7.2 The Six Recurring Mistake Patterns

7.2.1 Mistake 1: Ignoring Endogenous Price Feedback

Pattern. The claim says "X type of agent saves more, so capital is higher." This ignores that higher capital lowers the rate of return, which in turn affects saving choices.

When to deploy: Whenever the claim follows the chain "parameter shifts $\Rightarrow$ saving shifts $\Rightarrow$ aggregate stock shifts" without mentioning prices.

Past instance: 2016 Q2. *Claim.* "Economist A argues that the economy with z_s^+ has a higher skill premium." *Critique.* The skill premium is determined by relative marginal products. With more efficient skilled workers, effective skilled labor supply rises, lowering the skilled wage per efficiency unit. Whether the per-worker skill premium rises or falls depends on the elasticity of substitution and on the relative magnitudes. With Cobb–Douglas $L = S^{\phi}U^{1-\phi}$, the per-efficiency-unit skill wage falls but the per-worker skill wage may still rise (if z_s rises faster than the wage falls). **Need to derive carefully.**

Past instance: 2021 Q1d. *Claim.* "Economist Y argues that rental rate of capital in steady state in A will be lower." *Critique.* The setup: economies A and B differ in efficiency h of secondary earners ($h^A > h^B$). In A, more secondary earners participate (cutoff q^* is higher in h, established in part b), labor supply is larger. In Cobb–Douglas, $r = \alpha(K/L)^{\alpha-1}$. Whether r is lower in A depends on whether K/L is higher or lower in A. With log + CD, both K and L scale, so check the steady-state $k^* = K/L$ formula carefully. **Likely wrong if K scales less than L.** Resolution: derive the steady state and compare.

7.2.2 Mistake 2: Confusing Partial and General Equilibrium

Pattern. The claim assumes one variable shifts holding others constant, but in equilibrium they all move together.

Past instance: 2022 Q2c. *Claim.* "In steady state, GDP per worker will be higher in A. Type-L individuals save more, so the rate of return on capital will be lower in economy A." *Critique.* Type-L individuals (lower valuation of home goods) save more because they consume less of the home good. But aggregate saving depends on the population mix *and* on equilibrium prices. The first claim (GDP per worker higher in A) is backwards: with more type-L agents who care less about home goods, market hours are higher; but *output per worker* depends on the capital-labor ratio, not on hours per se. The second claim (return on capital lower in A) requires checking whether the higher saving translates into a higher K/L, which it does in log + CD. So the second claim is right, but the first claim is suspect.

Past instance: 2024 Q2c. *Claim.* "Economist GV argues that hours worked in the market will remain constant since the economy is on a balanced-growth path at $t_0 - 1$." *Critique.* A permanent rise in A (home productivity) shifts the marginal trade-off between market and home work. On the BGP, market hours can be constant only if the home and market goods grow at the same rate *or* preferences are over the right composite. With CRRA on home goods and log on market consumption, the BGP property may break. Need to check. Generally: a one-time shock to a parameter does *not* preserve constant market hours unless the preferences have very specific structure.

7.2.3 Mistake 3: Ignoring the Extensive (Participation) Margin

Pattern. The claim discusses intensive-margin effects, ignoring that the cutoff/threshold itself shifts.

Past instance: 2021 Q1d (related). *Claim.* "In steady state, output and the participation rate of secondary earners will be higher in A." *Critique.* Higher h raises the cutoff q^*, so more households cross the threshold to two-earner. Participation rate rises (correct for A). Output is higher because effective labor is higher. *Verify the participation channel is not double-counted in your math.*

7.2.4 Mistake 4: Confusing Planner and Competitive Allocations

Pattern. The claim asserts inefficiency or efficiency without considering whether the externality / market failure is present.

Past instance: 2020 Q1e. *Claim.* "In a competitive equilibrium for this economy, individuals work *too much*, and such equilibrium is inefficient." *Critique.* The model has an externality: individuals value others' leisure, but in a competitive equilibrium, no one internalizes this. So leisure is a public good, under-provided. Hours worked are *too high* relative to the planner's choice. The claim is correct *provided* we interpret "inefficient" in a Pareto sense w.r.t. the planner's allocation. Confirm the claim with the externality reasoning.

Past instance: 2021 Q2d. *Claim.* "Economist A argues that due to distortionary taxes, the competitive equilibrium is inefficient." *Critique.* The tax is on income = dividends + asset value, with lump-sum rebates. In a representative-agent economy with no government spending and lump-sum rebates, the tax has no real effect on the allocation: the agent receives back what she pays. The price level adjusts but the real allocation is unchanged. **The claim is wrong; this tax is non-distortionary in the rep. agent setup.**

7.2.5 Mistake 5: Mixing Levels and Growth Rates

Pattern. The claim discusses long-run effects on *levels* when only *growth rates* can change, or vice versa.

Past instance: 2017 Q3d. *Claim.* "Higher rates of taxation are needed to finance public infrastructure and therefore, increase the long-run level of investment in private capital." *Critique.* Higher τ has two effects. (i) Reduces private disposable income (negative on private saving / capital). (ii) Increases public infrastructure z, which raises the marginal product of private capital (positive). The net effect depends on the elasticity of capital w.r.t. infrastructure vs. the income effect on saving. **There is an interior optimal τ.** The claim is too strong; the relationship is hump-shaped.

7.2.6 Mistake 6: Conflating Individual and Aggregate

Pattern. The claim attributes an aggregate property to an individual or vice versa.

Past instance: 2019 Q2d. *Claim.* "Because the labor endowment of type-C agents grows over their lives, even when there is no growth in population or labor efficiency, the steady state equilibrium of this economy is not dynamically efficient." *Critique.* Dynamic efficiency depends on whether $r > n + g$ in steady state, where g is aggregate productivity growth and n is population growth. The fact that an *individual's* endowment grows over the life cycle does not affect aggregate growth (no aggregate g here). With $n = 0$ and $g = 0$, dynamic efficiency requires $r > 0$. With log + CD, the steady-state r depends on β, α, ϕ, and may exceed zero. **The claim is wrong**; the life-cycle profile of individuals does not introduce aggregate growth.

Past instance: 2017 Q1d. *Claim.* About persistence affecting price changes when z_t changes. *Critique.* Higher persistence makes the dividend more predictable, so its capitalized value moves more sharply with the realization. The amount by which $p_1 - p_2$ varies with z_t increases with persistence. (Not a Mistake 6 instance per se, but tests a clean GE-vs-dynamics distinction.)

7.3 Mapped Instances Catalog

Year/Q	**Claim**	**Mistake type**	**Verdict**
2016 Q2	z_s^+ economy has higher skill premium	1 (price feedback)	Ambiguous; depends on ϕ
2017 Q1d	Persistence raises asset-price volatility	5 (dynamics)	Correct (right reason needed)
2017 Q3d	Higher τ raises long-run private capital	5 (level vs. growth)	Wrong (hump-shaped)
2019 Q1e	Stock prices in country model exceed those with linear utility/no bond	4 (GE)	Carefully derive
2019 Q2d	Type-C life-cycle implies dynamic inefficiency	6 (individual vs. aggregate)	Wrong
2020 Q1e	Hours too high in CE due to public-leisure externality	4 (externality)	Correct
2021 Q1d	In economy A, output higher, participation higher (X), and rental rate of capital lower (Y)	1 (price feedback)	Mostly correct; verify carefully
2021 Q2d	CE inefficient due to distortionary taxes	4 (planner vs. CE)	Wrong (lump-sum rebate)
2022 Q2c	Higher π (more L-types) raises GDP per worker; lower r	1 + 6	Mostly wrong

Year/Q	Claim	Mistake type	Verdict
2023 Q1d	Asymmetric δ raises/lowers interest rate (vs. $\delta^A = \delta^B$)	1 + 6	Lower interest rate (Compute)
2024 Q2c	GV: market hours constant due to BGP	2 (BGP fragility)	Wrong

7.4 Template Answers

When the claim is wrong: "Economist X is incorrect. She has not accounted for [missing channel]. In equilibrium, [missing variable] adjusts so that [opposite or different conclusion]. Specifically, [show the math]."

When the claim is right but for the wrong reason: "Economist X reaches the correct conclusion, but the reasoning is incomplete. The right argument is [correct mechanism]. The mechanism she suggests would be [partial story], but in equilibrium [also include other channel]."

When the claim is ambiguous: "The answer depends on [parameter / functional form / elasticity]. If [condition A], her conclusion holds because [reason]. If [condition B], the opposite holds because [reason]. To determine which case applies, we need [specific computation]."

8 Part VII — Common Pitfalls and Notation

This section is your defense against the silly mistakes that cost the most points. Read it before each practice exam.

8.1 Time Conventions

Time Subscripts

- **Capital convention:** K_t is the capital stock available for production *in* period t. It was decided at the end of period $t-1$. Investment in period t becomes K_{t+1}.
- **Wage and return convention:** w_t, r_t are factor prices *in period* t. They equal the marginal products of inputs used in t.
- **OLG cohort:** "Born at t" = young at t, old at $t+1$. The young of t become the old of $t+1$.
- **Asset return:** when paying for a bond at t that pays at $t+1$, the gross return between t and $t+1$ is $1+r_{t+1}$. The Euler equation links $u'(c_t)$ to $\beta(1+r_{t+1})u'(c_{t+1})$.
- **Search timing:** In 2020 Q2 / 2022 Q3, the convention is: at the start of period t, the worker is in some state (employed at w, unemployed eligible, etc.). She decides borrowing/saving and accepts/rejects the offer. The offer accepted today starts paying *next* period. *Verify the convention from the question — it varies.*

8.2 Aggregate vs. Individual Variables

Capital Letters Matter

- Capital K_t vs. per-worker $k_t = K_t/L_t$. The Cobb–Douglas wage is $w_t = (1-\alpha)k_t^\alpha$ in per-worker terms; in aggregate terms it is $W_t L_t$ with $W_t = (1-\alpha)(K_t/L_t)^\alpha$.
- Aggregate consumption C_t vs. per-capita c_t. The market-clearing condition is $C_t + I_t = Y_t$, but per-capita representations divide both sides by L_t.
- Big K vs. little k in heterogeneous-agent RCE: little k is the individual choice; big K is the aggregate. The agent's value depends on (k, K). In equilibrium $K = \int k_i \, \mathrm{d}i$.

8.3 Notational Multipurpose Variables

The same letter often means different things across problems. Pay attention to context:

Symbol	Meaning(s) on past exams	Disambiguation
z	Dividend (asset pricing); productivity / TFP shock; entrepreneurial idea (2024 Q3); skill (2019 Q2: z^i); endowment (2023 Q1); leisure share (2020 Q1: not z but watch for similar)	Read the problem; in OLG/heterogeneity, z^i is endowment or skill; in tree models, z is dividend
δ	Depreciation rate of capital (most exams); mortality / death probability (2018 Q2); endowment loss when old (2023 Q1)	"Population grows at δ"? Then mortality. "Capital depreciates at δ"? Then standard.
ϕ	Production share / fraction (2016 Q2: $L = S^{\phi}U^{1-\phi}$); Cobb–Douglas exponent; productivity constant (2017 Q2: Φ); disutility of work (2019 Q3); private bond price coefficient	Distinguish from Φ (capital phi), often a constant in production.
γ	Taste parameter (2022 Q2 home goods; 2024 Q3 switching cost); skill exponent (2021 Q3); survival probability (2022 Q1: γ_S, γ_R); rate of return to home production (2017 Q3: λ similar)	Survival in projects, taste in OLG, exponent in skill.
ρ	Job separation probability (2020 Q2); disaster severity (2024 Q1: capital surviving fraction); persistence parameter (2017 Q1)	Context determines.
τ	Tax rate (most exams)	Always tax rate. Subscripts indicate tax type (τ^l, τ^k).
h	Hours of work; secondary earner efficiency (2021 Q1)	Sometimes h is hours at home (2024 Q2), sometimes secondary earner endowment.
b	Unemployment benefit; bond holdings (2023 Q2)	Context; bond holdings appear when κb in utility.
Φ vs. ϕ	Φ is the productivity constant in 2017 Q2; ϕ is the share	Capital vs. lowercase.

8.4 Sign Conventions in Derivations

FOC Sign Errors

- When k' is the choice and $c = f(k) - k'$, $\partial c/\partial k' = -1$. So FOC has $-u'(c) + \beta V'(k') = 0$, i.e. $u'(c) = \beta V'(k')$.
- When a' is the choice in asset pricing, $c = (p+d)a - pa'$, so $\partial c/\partial a' = -p$. FOC: $-pu'(c) + \beta \mathbb{E} V_a(a', z') = 0$.
- In OLG (sequential), the saving choice s enters with -1 in the young's budget and $+R_{t+1}$ in the old's. The sign of the Lagrangian on the lifetime budget should be $+\lambda$, not $-\lambda$.

8.5 Common Mathematical Errors

Math Pitfalls

- **Geometric series.** $\sum_{t=0}^{\infty} \beta^t = 1/(1-\beta)$, not $\beta/(1-\beta)$. The latter is $\sum_{t=1}^{\infty} \beta^t$.
- **Log of geometric:** $\ln \sum \beta^t \cdot \text{const} \neq \sum \beta^t \ln(\text{const})$. Be careful when computing lifetime utility.
- **Cobb–Douglas factor income shares:** with $Y = K^{\alpha} L^{1-\alpha}$, capital share is α, labor share is $1-\alpha$. Always.
- **Inada conditions:** $\lim_{c \to 0} u'(c) = +\infty$ rules out interior corner where $c = 0$. $\lim_{c \to \infty} u'(c) = 0$ rules out infinite consumption. State both when invoking Inada.
- **Discounting of streams:** an annuity paying b forever from $t = 0$ has present value $b/(1-\beta)$ at $t = 0$. From $t = 1$ it has PV $\beta b/(1-\beta)$ at $t = 0$.

- **CRRA:** $u(c) = c^{1-\sigma}/(1-\sigma)$ for $\sigma \neq 1$, and $u(c) = \ln c$ for $\sigma = 1$ (the limit). Use the right one.

8.6 Equilibrium-Definition Pitfalls

Equilibrium Definition Errors

- **Forgetting market clearing.** Always include the market-clearing condition in the equilibrium definition. Goods, capital, labor, asset, and (in OLG) the bond/loan market.
- **Forgetting the consistency condition.** In RCE, the aggregate law of motion must be consistent with the policy functions of individuals. Skipping this loses 1–2 points reliably.
- **Wrong equilibrium objects.** For asset pricing, the price function $p(z)$ is part of the equilibrium. For RCE with heterogeneity, the cross-sectional distribution is part of the equilibrium state.
- **Confusing prices with allocations.** Prices and allocations both must satisfy the equilibrium conditions. Don't list one without the other.
- **Initial conditions.** For OLG, list initial K_0 and the initial old generation's endowment / wealth. For Lucas tree, list initial asset endowment per agent.

8.7 Closed-Form Pitfalls

Closed-Form Errors

- **Saving rate confusion:** in standard log + CD growth, the saving rate is $\beta\theta$ (planner) or $\beta\theta$ (CE without taxes; same). Don't confuse with the OLG saving rate $\beta/(1+\beta)$.
- **Steady-state form:** the steady state of $k_{t+1} = \phi k_t^\alpha$ is $k^* = \phi^{1/(1-\alpha)}$. **Not** ϕ or $\phi^{1/\alpha}$.
- **Forgetting the population growth factor:** $k_{t+1} = s_t/(1+n)$ in OLG with population growth, not $k_{t+1} = s_t$.
- **Wrong return formula:** with $\delta = 1$, $R = \alpha k^{\alpha-1}$ (gross). With $\delta < 1$, $R = 1 - \delta + \alpha k^{\alpha-1}$.

9 Part VIII — Annotated Past-Exam Index

For each problem in the 9 sample exams, this index provides:

- **Framework:** Topic + Topic-specific variant
- **Tests:** What skill or concept is being assessed
- **Trick/Punchline:** The non-obvious insight or move
- **Links:** Sections of this playbook to consult

9.1 June 2016 (Choose 3 of 4)

Problem 1: Standard growth with N heterogeneous households. Framework: Planner growth with multiple agents, log utility, Cobb–Douglas.
Tests: Understanding that with log utility, individual consumption is a constant fraction of aggregate consumption (negishi-like aggregation). Definition of Arrow–Debreu equilibrium. Whether equilibrium aggregate path depends on initial distribution.
Punchline: With log utility, the aggregate dynamics are independent of the wealth distribution. The CE matches the planner because there are no externalities.
Links: Section 4.1 (log+CD), Section 3.3.3 (AD definition).

Problem 2: OLG with skilled and unskilled labor. Framework: Two-period OLG with two labor types, log utility, $L = S^{\phi}U^{1-\phi}$.
Tests: Equilibrium definition with heterogeneity. Steady-state comparison across $z_s^+ > z_s^-$. "Economist X" on skill premium.
Punchline: Higher z_s raises effective skilled labor, lowering the per-efficiency-unit wage; the per-worker premium goes the other way. Need to derive the wage formula explicitly.
Links: Section 6.1, Section 7 Mistake 1.

Problem 3: Asset pricing with two trees, log utility, iid binary first-tree dividend. Framework: Lucas tree, two trees, second tree has constant dividend = 1.
Tests: Closed-form pricing with log; recognition that consumption is the sum of dividends.
Punchline: p_2 may not equal $\beta/(1-\beta)$ because consumption fluctuates with z_1. Carefully derive both prices.
Links: Section 6.2, Recipe 4 (variant).

Problem 4: Ramsey taxation with capital. Framework: Standard Ramsey, τ^l, τ^k. Show $\tau^k = 0$ for $t \geq 1$.
Tests: A–D equilibrium definition, implementability constraint, classic Chamley–Judd result.
Links: Section 6.4.

9.2 June 2017 (Choose 3 of 4)

Problem 1: Two trees with perfectly negatively correlated dividends (z and $1-z$). Framework: Lucas asset pricing. Shows $p_1^*(z) + p_2^*(z) = a$ constant.
Tests: Recognition that $c = z_1 + z_2 = 1$ is constant in equilibrium, so $u'(c)$ is constant. Then prices linear in dividend states.
Punchline: $a = \beta/(1-\beta)$ (the consumption value of the entire economy). Persistence affects how

$p_1 - p_2$ varies with z_t.
Links: Section 6.2, Recipe 4 variant.

Problem 2: Endogenous growth with capital externality. **Framework:** $A_t = \int k_{i,t}\, \mathrm{d}i$. Equilibrium vs. planner.
Tests: Recognition that private return is $\theta\Phi$ (mp w.r.t. own capital), planner return is Φ (mp w.r.t. aggregate). For intermediate Φ, equilibrium growth is negative while planner is positive.
Punchline: Optimal policy: subsidy to capital combined with offsetting labor tax. Wedge is $1/\theta$.
Links: Section 6.4.

Problem 3: OLG with public infrastructure. **Framework:** $y = z^\lambda k^\alpha l^{1-\alpha}$, labor income tax τ funds infrastructure.
Tests: Steady-state K/L; "Economist X" on hump-shaped relation between τ and capital.
Punchline: Two effects of τ: lowers private saving directly (through $w(1-\tau)$); raises marginal product of capital indirectly (through higher z). Hump-shaped optimum.
Links: Section 6.1, Section 7 Mistake 5.

Problem 4: Optimal labor income taxation, no capital, quasi-linear utility. **Framework:** $u = c - l^{1+\varphi}/(1+\varphi)$. Compare balanced budget vs. constant tax rates.
Tests: Implementability constraint without capital. Recognition that DWL is convex in τ, so smoothing is optimal.
Punchline: Constant tax rates dominate balanced budget by Jensen's inequality applied to the convex DWL function.
Links: Section 6.4.

9.3 May 2018

Problem 1: Asset pricing with time-varying net supply ($\bar{y}_t$ alternating). **Framework:** Single tree, but external investors absorb part of supply; net supply alternates between 1 (odd) and 2 (even).
Tests: Aggregate state is $\bar{y}$; per-capita consumption $c_t = \bar{y}_t \cdot z$ alternates. Linear vs. concave u comparison.
Punchline: With linear u, prices equal expected discounted dividends, same in both states. With concave u, prices differ because $u'(c)$ differs across states (consumption smoothing motive).
Links: Section 6.2, Section 3.4.

Problem 2: Three-period OLG with mortality and TFP growth. **Framework:** Three life stages; survival probability $\beta = 1 - \delta$; $A_t = (1+g)^t$; Cobb–Douglas.
Tests: Three-period lifetime budget. Aggregate savings as sum across cohorts. Steady-state growth rate of K/L. Saving rate vs. δ (mortality).
Punchline: On BGP, K/L grows at rate g (BGP property). The saving rate as a function of δ: higher δ (more death) lowers expected lifetime utility weight on later periods, reducing saving.
Links: Section 6.1.

Problem 3: Resource extraction (lumber) with growth. **Framework:** $x_{t+1} = (x_t - l_t)(1+g)$; $c_t = f(l_t)$, log utility, power production.
Tests: Custom law of motion; Euler equation derivation; closed-form with log + power; trade between two agents with different g.

Punchline: Steady-state requires $\beta(1+g)f'(l) = 1$ in some form. With log + $f(l) = l^\gamma$, the consumption profile is constant iff $\beta(1+g) = 1$. Trade: agent with higher g has higher resource growth, consumes more in future, hence borrows in period 0.
Links: Section 6.5.

9.4 June 2019

Problem 1: Two-country asset pricing with international bond. **Framework:** Each country has its own tree (only its residents trade); a single international bond links the two.
Tests: RCE in heterogeneous-country setting. Stationary equilibrium where consumption is constant when dividends alternate.
Punchline: Constant consumption $\Rightarrow$ $u'(c)$ pulls out of the SDF. Bond price equal across countries (no-arbitrage). Asset prices reflect the residents' hand-to-mouth aggregate consumption.
Links: Section 6.2.

Problem 2: OLG with college / non-college labor profile. **Framework:** Type C has labor profile (1, 1+g); type NC has (1, 1). Log utility.
Tests: Saving decisions, possibility of borrowing, steady-state K/L, "Economist X" on dynamic efficiency.
Punchline: Type C has rising income profile, may want to borrow when young to smooth consumption. Whether they actually do depends on borrowing constraint. Higher ϕ (more type C) raises future labor supply, but per-worker effects on K/L depend on saving behavior. Dynamic efficiency: requires comparing r to aggregate growth rate; individual life-cycle profile is irrelevant.
Links: Section 6.1, Section 7 Mistake 6.

Problem 3: Irreversible employment choice (Galina-style). **Framework:** Choose once at $t = 0$ whether to work forever or never. Fixed wage, savings, $a_t \geq 0$.
Tests: Cutoff rule in a_0. Proof that consumption when working exceeds consumption when not working.
Punchline: For low a_0, work; for high a_0, don't (the disutility ϕ is fixed, but the marginal utility of additional income falls with wealth). Cutoff $\bar{a}_0$ exists.
Links: Section 5.1, Section 5.6.

9.5 June 2020

Problem 1: Optimal growth with public-leisure externality. **Framework:** $\phi(l, \bar{l})$ in utility; planner internalizes $\bar{l} = l$. Closed-form with log + log + log + CD + $\delta = 1$.
Tests: Recursive planner setup. FOC + Envelope. Steady-state K/L same as standard model. "Economist X" on inefficiency: yes, hours too high in CE.
Punchline: The externality on leisure means competitive equilibrium under-provides leisure (over-supplies labor), so hours are too high. The K/L ratio is invariant because leisure does not enter the marginal product of capital condition in the long run.
Links: Section 4.1, Section 7 Mistake 4.

Problem 2: McCall search with assets. **Framework:** Worker has bond market (a). Reservation wage $w^*(a)$.
Tests: Stationary RCE definition. Proof that $w^*(a)$ is increasing in a when $c_e(a) > c_u(a)$.
Punchline: Use envelope on both value functions; the indifference condition links derivatives. The

sign of $\partial w^*/\partial a$ comes from the sign of $u'(c_u) - u'(c_e)$.
Links: Section 5.2, Section 6.3.

Problem 3: Two-period OLG with social security. **Framework:** $d_t = \tau w_t$, $p_t = (1+n)\tau w_t$. Standard log + CD + $\delta = 1$.
Tests: Pension formula, saving function with social security, effect of τ on K/L.
Punchline: τ reduces private saving (less disposable income, plus pension wealth substitutes). K/L falls in τ.
Links: Recipe 3.

9.6 June 2021

Problem 1: OLG with two-person households and joint work. **Framework:** Each household has two members (one with full endowment, one with $h \leq 1$). Disutility q if both work; q heterogeneous with cdf G.
Tests: Cutoff q^* in heterogeneity. Show q^* increases in h. Steady-state K/L. "Economist X" on output, participation, and rental rate.
Punchline: Higher h raises the relative wage of the secondary earner, raising the cutoff (more participation). The effect on K/L depends on whether saving (proportional to wage income) rises faster than labor.
Links: Section 5.1, Section 6.1.

Problem 2: Lucas tree with proportional income tax. **Framework:** Tax on dividends + asset value, lump-sum rebate.
Tests: Closed-form pricing with log + tax. Show prices decreasing in τ. "Economist X" on inefficiency.
Punchline: $p(z) = \beta(1-\tau)/(1-\beta(1-\tau))z$ is decreasing in τ. But the equilibrium allocation is unaffected (all consumption is just z); rebate offsets the tax. Hence, no inefficiency despite distortion in nominal terms.
Links: Section 6.2, Section 7 Mistake 4, Section 5.4.

Problem 3: Skill accumulation, no asset markets, log utility. **Framework:** $s_{t+1} = (1-\delta)s_t + As_t^{1-\gamma}i_t^{\gamma}$, $c_t = ws_t - i_t$.
Tests: Recursive setup. Euler equation. Guess-and-verify with log: investment linear in skill. Effect of δ on investment.
Punchline: Investment $i_t = \xi s_t$ where ξ depends on parameters. ξ is decreasing in δ (higher depreciation reduces marginal benefit of investment).
Links: Section 5.3, Section 6.5.

9.7 June 2022

Problem 1: Project choice (Galina-style). **Framework:** Safe vs. Risky project, different A_i and γ_i. Choose once at $t = 0$. Failure $\Rightarrow$ benefit b forever.
Tests: Recursive setup. Cutoff $\bar{k}_0$. Counter to claim about consumption at $t = 0$ being higher with project S. Modification with per-period choice.
Punchline: For low k_0, the consumption-insurance benefit b dominates and R is preferred. For high k_0, survival probability dominates and S is preferred. Initial consumption argument: with log utility and AK-like structure, c_0 is proportional to k_0, but the proportionality constant differs across projects. The claim that c_0 is higher with S than R depends on parameters; rule it out by a careful Bellman

comparison.
Links: Section 5.1, Section 6.5.

Problem 2: OLG with home production and heterogeneous taste. **Framework:** Home good $h = G(l) = l^\lambda$. Two types L, H with $\gamma_L < \gamma_H$.
Tests: Law of motion for K/L; steady-state existence; comparative statics across π.
Punchline: L types substitute market for home (work more); H types prefer home goods (save less due to lower wage income). Whether r is lower in A (more L types) depends on the trade-off in saving and labor supply.
Links: Section 6.1, Section 7 Mistake 1.

Problem 3: McCall search with temporary unemployment compensation. **Framework:** Workers eligible for b only first period; can quit; reservation wages w_e^*, w_1^*, w_+^*.
Tests: Three-state Bellman. Reservation wage form. Ordering of three reservation wages. Difference $w_1^* - w_+^*$.
Punchline: $w_e^* < w_+^* < w_1^*$. Workers with longer past unemployment have lower reservation wages, so they accept more jobs but are also more likely (subsequently) to quit when conditions are unfavorable.
Links: Section 6.3, Section 5.2.

9.8 June 2023

Problem 1: OLG without capital, two endowment types. **Framework:** Pure consumption-loan OLG. Type A and B with different endowments and δ^i (fraction of endowment lost in old age).
Tests: Equilibrium interest rate. Show $r < 1/\beta$. No-trade if $\delta^A = \delta^B$. Effect of asymmetry Δ on r.
Punchline: With identical types, no gains from trade $\Rightarrow$ no trade. Asymmetry Δ creates desire for cross-type insurance, raising/lowering r depending on which type is the lender.
Links: Section 6.1.

Problem 2: Lucas tree with bond holdings in utility. **Framework:** $u(c + \kappa b)$. Stock + risk-free bond.
Tests: Two Euler equations. Closed-form with log. Equity premium.
Punchline: The bond's direct utility from κ raises its demand, lowering its risk-free rate. The stock's expected return is unchanged. Hence equity premium increases with κ.
Links: Section 6.2, Recipe 4.

Problem 3: AK growth with two agents, joint vs. alternating management. **Framework:** (a) Independent management: each agent solves separate AK problem. (b) Alternating: each agent withdraws goods, growth at A for one period only; strategic interaction.
Tests: Standard AK + log; conjecture symmetric linear policies in alternating case; show aggregate stock grows slower under alternating.
Punchline: Under alternating, each agent free-rides (depletes the shared pool more aggressively because the other will do the same next period). Aggregate growth rate is lower.
Links: Section 5.3, Section 6.5.

9.9 June 2024

Problem 1: AK growth with disasters. **Framework:** Probability q of disaster wiping out $(1-\rho)$ of capital. CRRA utility.
Tests: Recursive planner setup. Euler equation. Closed-form saving rule. Effect of q and ρ on saving

(depends on $\sigma \gtrless 1$).
Punchline: Two competing effects: precautionary saving (raises saving) vs. lower expected return (lowers saving). Sign depends on σ.
Links: Recipe 6, Section 6.5.

Problem 2: OLG with home production and TFP growth. Framework: Linear home production $c_2 = Ah$. Cobb–Douglas market production. CRRA utility on home goods, log on market.
Tests: BGP analysis. Whether market hours are constant. Effect of permanent rise in A (home productivity) on rate of return.
Punchline: Permanent rise in A shifts agents toward home production, lowering market labor supply. Rate of return falls (with less effective labor in market production, K/L rises). The claim that hours stay constant on BGP is wrong: BGP requires specific elasticities.
Links: Section 6.1, Section 7 Mistake 2.

Problem 3: Continuum of agents with worker/entrepreneur choice. Framework: Markov idiosyncratic entrepreneur quality z. Government taxes profits to finance G. Switching cost γ.
Tests: RCE definition with heterogeneous-agent occupational choice. Effect of unexpected G increase on wages.
Punchline: Higher G requires higher tax revenue $\Rightarrow$ higher tax rate on profits. Lower after-tax return to entrepreneurship $\Rightarrow$ workers' supply rises (some entrepreneurs switch to workers). Wage falls.
Links: Section 3.3.1, Section 6.5.

Appendix: Day-of-Exam Cheat Sheet

This single page is what you should re-read in the 30 minutes before the exam, after you have closed all other materials.

The Five Steps

1. **Set up:** state, controls, constraints.
2. **FOC + Envelope ⇒ Euler.**
3. **Equilibrium definition:** prices, allocations, optimization, market clearing, consistency.
4. **Closed form:** log + CD + $\delta = 1$ + iid.
5. **Comparative statics or claim evaluation.**

Three Equilibrium Templates (Memorize)

RCE: $\{V, g, p, \Gamma\}$ such that (i) V, g solve Bellman given p, Γ; (ii) markets clear; (iii) Γ consistent with g.
Sequential CE: {allocations, prices} such that (i) household optimization; (ii) firm profit max; (iii) markets clear; (iv) initial conditions.
Arrow–Debreu: {allocations, p_t, w_t, r_t} such that (i) date-0 budget binding; (ii) firms profit max; (iii) gov't BC in PV; (iv) markets clear.

Closed-Form Lookup

Standard log + CD growth:	$c = (1-\beta\theta)k^\theta$, save rate $\beta\theta$
Log OLG (CD, $\delta = 1$):	$s = \frac{\beta}{1+\beta} w$
Lucas tree, log:	$p(z) = \frac{\beta}{1-\beta} z$
Log AK growth:	$k' = \beta A k$, $c = (1-\beta)Ak$
McCall reservation wage:	$w^* = b + \frac{\beta}{1-\beta} \int_{w^*}^{\bar{w}} (1-F)\, \mathrm{d}w$

Three Reflexes

1. **If you see "log + CD + $\delta = 1$"**: try guess-and-verify $V(k) = a + b \ln k$, expect linear policies.
2. **If you see "Show that there is a cutoff"**: monotonicity + IVT in two value functions; plot intuition.
3. **If you see "Economist X claims"**: identify the missing GE channel, write 3–4 sentences.

Common Pitfalls Quick-Check

- Aggregate vs. per-worker: use K_t/L_t when in doubt.
- OLG cohort dating: capital saved at t is used at $t+1$.
- Pension formula: $p_t = (1+n)\tau w_t$ in PAYG.
- Population growth: $k_{t+1} = s_t/(1+n)$, not s_t.
- Steady state of $k = \phi k^\alpha$ is $k^* = \phi^{1/(1-\alpha)}$.

- Geometric series from $t = 0$: $1/(1-\beta)$. From $t = 1$: $\beta/(1-\beta)$.
- RCE definition must include consistency condition.
- "Claim is wrong" is the modal answer. Look for missing GE channel.

Time Allocation

- First 5 minutes: read all problems, identify categories, allocate time.
- 1 hour per problem (3-problem exam) or 45 minutes (4-problem exam, choose 3).
- Reserve 10 minutes per problem for proofreading and equilibrium-definition cleanup.

If You Are Stuck

1. Write the Bellman equation. Even if you can't solve it, you'll get partial credit.
2. Try the simplest closed-form trick (log + CD + $\delta = 1$) even if not stated; you'll often see it works.
3. For "Economist X": just say what *should* happen (the right comparative static), even if you can't formally prove it.
4. Move on. Better to have 90% of three problems than 100% of one.

Trust the patterns. Trust the templates. Take your time.